

-2 / +14 lines of 13 · 1 parallel region(s) added · 4 of 12 candidate(s) kept · RESTRUCTURING (13 code line(s) added, 2 removed) · restructuring level 0

BEFORE

```
1  #include "data.h"
2
3  real_t kernel_k53(void)
4  {
5      for (int nl = 0; nl < iterations; nl++) {
6
7          for (long i = 1; i < LEN_1D; i++) {
8              u[ju[i]] += v[kv[i]] * c[i];
9              v[jv[i]] = u[ku[i]] * d[i] + c[i];
10
11          }
12      }
13  }
```

AFTER

```
1  #include "data.h"
2
3  real_t kernel_k53(void)
4  {
5      for (int nl = 0; nl < iterations; nl++) {
6          real_t u_snapshot[LEN_1D], c_snapshot[LEN_1D], d_snapshot[LEN_1D];
7          for (long i = 0; i < LEN_1D; i++) {
8              u_snapshot[i] = u[i];
9          }
10         for (long i = 1; i < LEN_1D; i++) {
11             c_snapshot[i] = c[i];
12             d_snapshot[i] = d[i];
13         }
14         for (long i = 1; i < LEN_1D; i++) {
15             u_snapshot[ju[i]] += v[kv[i]] * c_snapshot[i];
16             v[jv[i]] = u_snapshot[ku[i]] * d_snapshot[i] + c_snapshot[i];
17         }
18         #pragma omp parallel for firstprivate(u_snapshot)
19         for (long i = 0; i < LEN_1D; i++) {
20             u[i] = u_snapshot[i];
21         }
22         dummy(a, b, c, d, e);
23     }
24     return (real_t)0;
25 }
```

